

	CIRCULATORY HISTOLOGY	Question time (seconds): →	
۱	In arteriole:		
	۱	Tunica media contain ۲-۳ muscle layers	
	۲	Distinct internal elastic lamina	
	۳	Elastic fibers are high amount	
	۴	Tunica externa is thick	
Source:			
Description:			

	CIRCULATORY HISTOLOGY	Question time (seconds): →	
۲	In the normal histological structure of the different types of capillaries:		
	۱	Sinusoids have small pores	
	۲	Endothelial cell has fenestrations in capillaries	
	۳	Discontinuous capillaries have tight junctions	
	۴	Thick basement membranes are in continuous capillaries	
Source:			
Description:			

	CIRCULATORY HISTOLOGY	Question time (seconds): →	
۳	Muscular artery include.....:		
	۱	Elastic fibers in Intima	
	۲	Thick tunica externa or adventitia	
	۳	Internal & External Elastic Lamina	
	۴	Tunica media have ۴ layers of muscle	
Source:			
Description:			

	CIRCULATORY HISTOLOGY	Question time (seconds): →	
۴	In the normal histological structure of veins:		
	۱	Adventitia have more elastic fibers	
	۲	Intima is thin	
	۳	Media has large muscle bundles	
	۴	Endothelial cells have tight junctions	
Source:			
Description:			

	CIRCULATORY HISTOLOGY	Question time (seconds): →	
۵	In thymic lymphocyte selection.....		
	۱	All Lymphoblasts express TCR	
	۲	In positive selection all cells express TCR	
	۳	The selection process occurred just in thymus cortex	
	۴	Cytotoxic cells express CD۴	
Source:			
Description:			

	CIRCULATORY HISTOLOGY	Question time (seconds): →	
۶	In microscopic structure of Spleen:		
	۱	Central artery is in white pulp.	
	۲	Red pulp located in PALS.	
	۳	Billrout's cord are clusters in white pulp.	
	۴	Stave cells are in close circulatory	
Source:			
Description:			

	CIRCULATORY HISTOLOGY	Question time (seconds): →	
۷	In lymphatic nodes.....:		
	۱	Subcapsular sinus was seen in medulla	
	۲	Medullary cords are extended from paracortex	
	۳	Paracortex have B lymphocyte	
	۴	Medullary sinuses extended inward from the capsule	
Source:			
Description:			

	CIRCULATORY HISTOLOGY	Question time (seconds): →	
۸	All of statements are correct about thymus EXCEPT:		
	۱	Medulla include continues capillary	
	۲	Hassall's body are remanence of Epithelioreticular cells	
	۳	Thymus lobules include Cortex and Medulla	
	۴	Hassall's body are basophil structure in Medulla	
Source:			
Description:			

	CIRCULATORY HISTOLOGY	Question time (seconds): →	
۹	Which of these are described lymphoid tissue?		
	۱	Lymphatic nodules include Cortex and Medulla	
	۲	lymphatic nodes found in clusters at connective tissue	
	۳	In Diffuse lymphatic tissue capsule is absent	
	۴	Spleen have oval-shaped sinuses	
Source:			
Description:			

	CIRCULATORY HISTOLOGY	Question time (seconds): →	
۱۰	Which of these tonsils don't have crypts:		
	۱	Lingual	
	۲	Tubal	
	۳	Palatine	
	۴	Pharyngeal	
Source:			
Description:			

		داخلی	
۱۱	۱. Administration of certain K channel antagonists causes the plateau of an action potential curve to be		
	a. prolonged	الف	
	b. diminished	ب	
	c. unchanged	ج	
	d. doesn't affect phase ۲ AP phase duration	د	
(Guyton p:115)			منبع:

۱۲	۲. Cardiac survival during cardiac transplantation is improved by perfusing donor hearts with cardioplegic solutions containing about ۲۰mM KCl. Why is this high potassium concentration helpful?		
	a. Elevation in ECF K repolarizes myocytes, activates Na & Ca channels, making the cells inexcitable and relaxed	الف	
	b. Elevation in ECF K depolarizes myocytes, activates Na & Ca channels, making the cells excitable and relaxed	ب	
	c. Elevation in ECF K repolarizes myocytes, inactivates Na & Ca channels, making the cells excitable and relaxed m	ج	
	d. Elevation in ECF K depolarizes myocytes, inactivates Na & Ca channels, making the cells inexcitable and relaxed	د	
(Guyton p:125)			منبع:

۳. Absolute refractory period in heart	۱۳
a. Lasts till cardiac contraction	الف
b. Longer than refractory period in skeletal musc	ب
c. A phase of cardiac cycle in which heart cannot be stimulated by any amount of stimulus	ج
d. All of the above	د
(Guyton p:116)	منبع:

۴. A ۵۵-year-old man has been diagnosed with Stokes-Adam syndrome. Two minutes after the syndrome starts active blockade of the cardiac impulse, which of the following is the pacemaker of the heart	۱۴
a. Sinus node.	الف
b. AV node.	ب
c. Purkinje fibers.	ج
d. Cardiac septum.	د
(Guyton p:131)	منبع:

۵. Which of the following phases of cardiac cycle follows immediately after the beginning of the QRS complex	۱۵
a. Isovolumic relaxation	الف
b. Ventricular ejection	ب
c. Atrial systole	ج
d. Isovolumic contraction	د
(Guyton p:118)	منبع:

۶. which of the following events occurs at the end of the period of ventricular ejection?	۱۶
a. AV valves close.	الف
b. Pulmonary valve closes.	ب
c. Aortic valve remains open.	ج
d. Aortic valve opens.	د
(Guyton p:120)	منبع:

۷. A ۳۰-year-old man has an ejection of ۰,۲۵ and an end systole volume of ۱۵۰ ml. what is his end-diastolic volume	۱ ۷
a. ۵۰ ml.	الف
b. ۱۰۰ ml.	ب
c. ۱۲۵ ml.	ج
d. ۲۰۰ ml.	د
(Guyton p:115)	منبع:

۱۸. Calcium is transported out of the cardiac cell by	۱۸
a. $\text{Na}^+/\text{Ca}^{++}$ -exchanger	الف
b. Ca^{++} -ATPase	ب
c. Na^+/K^+ -ATPase pump	ج
d. Passive diffusion down its concentration gradient	د
(Guyton p:117)	
منبع:	

۱۹. The T wave of the electrocardiogram begins to occur during	۱۹
a. early diastole	الف
b. isovolumetric contraction	ب
c. isovolumetric relaxation	ج
d. systolic ejection	د
(Guyton p:118)	
منبع:	

۲۰. The "a" wave of atrial pressure curve is due to	۲۰
a. atrial systole	الف
b. ventricular systole	ب
c. atrial diastole	ج
d. ventricular diastole	د
(Guyton p:118)	
منبع:	

۲۱. Ventricular systole is defined as the time between the	۲۱
a. closing of the aortic valve and the opening of the mitral valve	الف
b. closing of the mitral valve and the closing of the aortic valve	ب
c. opening and closing of the aortic valve	ج
d. opening of the aortic valve and the opening of the mitral valve	د
(Guyton p:118)	
منبع:	

۲۲. Which of the following are caused by acetylcholine?	۲۲
a. Hyperpolarization of SA node.	الف
b. Depolarization of AV node.	ب
c. Decrease permeability of the SA node to potassium ions.	ج
d. Increased heart rate.	د
(Guyton p:132)	
منبع:	

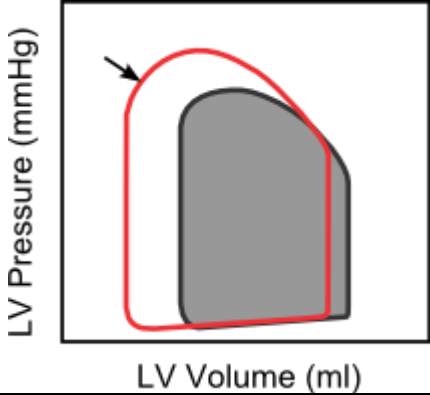
۲۳	۱۳. Sympathetic stimulation of the heart normally causes which of the following?
الف	a. Acetylcholine release at the sympathetic ending.
ب	b. Decrease heart rate.
ج	c. Decrease rate of conduction of the cardiac impulses.
د	d. Increase force of contraction of the ventricles.
منبع:	(Guyton p:115)

۲۴	۱۴. The maximal rate of pressure development within the left ventricle occurs
الف	a. During isovolumetric contraction
ب	b. During rapid ventricular ejection
ج	c. At peak left ventricular systolic pressure
د	d. During rapid ventricular filling
منبع:	(Guyton p:118)

۲۵	۱۵. Closure of the aortic and pulmonic valves produces which heart sound?
الف	a. S _۱
ب	b. S _۲
ج	c. S _۳
د	d. S _۴
منبع:	(Guyton p:121)

۲۶	۱۶. A ۷۰-year-old male had a following ECG during his annual physical examination. What is the expected duration of Q-T interval?
الف	a. ۰٫۱۲ sec
ب	b. ۰٫۱۶ sec.
ج	c. ۰٫۴۰ sec.
د	d. ۰٫۶۰ sec.
منبع:	(Guyton p:137)

۲۷	۱۷. When recording lead aVL on ECG, the positive electrode is the:
الف	a. Left arm.
ب	b. Left leg.
ج	c. Right leg.
د	d. Left arm & left leg
منبع:	(Guyton p:140)

۱۸. Compared to a normal left ventricular pressure-volume loop (grey), the red loop indicated by the arrow represents (Guyton p:118) 	۲۸
a. Increased preload, afterload, and inotropy	الف
b. Increased inotropy and afterload, and decreased preload	ب
c. Increased stroke volume with increased end-systolic volume	ج
d. Increased stroke volume and decreased ejection fraction	د
(Guyton p:122) منبع:	

۱۹. The last portion of the heart to depolarize is	۲۹
a. The entire outer surface of the apex	الف
b. The endocardial areas	ب
c. The inner surface of the apex	ت
d. Posterobasal portion of the left ventricle and the pulmonary conus	د
(Guyton p:146) منبع:	

۲۰. If the mean electrical axis is 90°, then the net amplitude of the QRS will be greatest in	۳۰
a. Lead I	الف
b. Lead III	ب
c. aVF	ج
d. aVL	د
(Guyton p:144) منبع:	

۲۱. In a Lead I ECG recording, the first small negative deflection of the QRS (the Q-wave) represents	۳۱
a. Atrial repolarization	الف
b. Depolarization of the interventricular septum from left to right	ب
c. A wave of depolarization traveling towards the left leg	ج
Ventricular repolarizationd.	د
(Guyton p:145) منبع:	

		۳۲
۲۲. Which statement IS NOT TRUE about erythropoietin?		
A small amount of erythropoietin is produced by non-kidney tissues.	الف	
Increases the speed of RBC development stages.	ب	
Hypoxia <u>directly</u> increases the synthesis of erythropoietin.	ج	
It causes the movement of proerythroblast production from stem hematopoietic cells.	د	

		۳۳
۲۳. Which option is <u>not</u> a characteristic of immature RBCs?		
short life span	الف	
different and inefficient shapes (spherical, oval-shaped)	ب	
Not carrying any amount of oxygen.	ج	
Existence of a fragile and vulnerable membrane.	د	
		منبع:

		۳۴
۲۴. In your opinion, when attacking the body, how do phagocytotic cells not attack their own cells?		
Through the formation of pseudopods	الف	
Through anti body attachment to the bacterial cell	ب	
Through phagosome formation	ج	
Through the development of inflammation.	د	
		منبع:

		۳۵
۲۵. In the trapping of viral particles by alveolar macrophages (specific macrophages reside in lung), which of the immune system is active?		
First line of defense	الف	
Second line of defense	ب	
Fourth line of defense	ج	
Margination	د	
		منبع:

		۳۶
۲۶. Which one does <u>not</u> increase in polycythemia?		
Blood viscosity	الف	
Blood resistance	ب	
Cardiac output	ج	
Blood hematocrit	د	
		منبع:

۳۷. Person A has vit B ^{۱۲} deficiency for ۵ years and person B has deficiency of vit B ^{۱۲} for ۵ months. Which of ۲ are susceptible for anemia?		۳۷
Only person A	الف	
Only person B	ب	
Both of them	ج	
None of them	د	
		منبع:

۳۸. Which of the following is common at both neutrophil and macrophage?		۳۸
Chemotaxis	الف	
Power of phagocytosis	ب	
Size of phagocytized particles	ج	
Amount of phagocytized particles	د	
		منبع:

۳۹. Which of the following is NOT correct about plateletes?		۳۹
They have no ability for ATP production	الف	
They have contraction filaments	ب	
They are able to become sticky	ج	
They are able to form pseudopods	د	
		منبع:

۴۰. The conversion of Prothrombin to Thrombin is done by		۴۰
t-PA	الف	
activated factor XII	ب	
prothrombin activator	ج	
fibrin	د	
		منبع:

۴۱. Production site of red blood cell during <u>childhood</u> is		۴۱
Yolk sac	الف	
Liver	ب	
Only Bone marrow of long bones (i.e. femur)	ج	
Bone marrow (vertebra)	د	
		منبع:

۳۲. What is the role of calcium ions in hemostasis?		۴۲
It is essential for all steps of blood coagulation in intrinsic pathway	الف	
It is essential for all steps of blood coagulation ONLY in extrinsic pathway	ب	
It is essential for anti-coagulation effects of vit K	ج	
It is not essential for ۲ first steps of intrinsic pathway	د	
		منبع:

۳۳. Which of the following cell exist in the bloodstream?		۴۳
Proerythroblast	الف	
Orthochromatic erythroblast	ب	
Reticulocyte	ج	
Tissue macrophages	د	
		منبع:

۳۴. About resistance in the circulation choose the right choice.		۴۴
PRU in small circulation is ۷-fold greater than big circulation	الف	
Increase in blood vessel radius strongly reduces its resistant	ب	
In the constant flow with increase flow resistance would increase	ج	
In the parallel system the total resistance are the sum of the each resistance	د	
		منبع:

۳۵. In which cases the pressure of right atrium would increase?		۴۵
Acute heart failure	الف	
Abdominal vain dilation	ب	
Increase in heart contractile pressure	ج	
Increase in PRU	د	
		منبع:

۳۶. In which situation embossment of neck veins are normal?		۴۶
Laying down	الف	
Sitting down	ب	
Standing up	ج	
Walking	د	
		منبع:

۴۷. In the regulation of local control of blood flow which of the following is the most important one?	۴۷
Tissue metabolites	الف
Growth factors	ب
Adenosine	ج
Lack of the O ₂	د
منبع:	

۴۸. which of the following vasodilators have stronger and long-lasting effects on vasodilatation?	۴۸
NO	الف
Angiotensin	ب
Histamine	ج
CGRP	د
منبع:	

۴۹. Increase in which of the following cause reduction in Reynolds No and change the blood stream to laminar flow?	۴۹
Diameter	الف
Density	ب
Viscosity	ج
Velocity	د
منبع:	

۵۰. In comparison between ۲ circulation which of the following indexes is equal in both of them?	۵۰
Resistance	الف
Flow	ب
Pressure	ج
Blood volume	د
منبع:	

۵۱. In the tension issue to the wall of the vessel, which one is correct?	۵۱
With increase in vessel parapet (wall) thickness, tension would increase	الف
With increase in its radius, the pressure to the wall decreases	ب
With reduction in blood velocity, parapet tension would increase	ج
Wall tension would increase directly with increased in blood pressure	د
منبع:	

۵۲	۴۲. Which of the following organs vessel doesn't have significant change in its blood flow with sympathetic stimulation ?
الف	Brain
ب	Skin
ج	Muscles
د	Kidney
منبع:	

۵۳	۴۳. In the artery and veins issue which one is correct.
الف	Arteries have highest volume of blood.
ب	Veins compliances are much more than arteries.
ج	Velocity of blood in veins is higher than arteries.
د	The cross section of arteries are more than veins.
منبع:	

۵۴	۴۴. About circulation pressure and its resistance, choose the correct one.
الف	MAP in the left ventricle is about ۹۰ mm Hg.
ب	PRU in both circulations are equal.
ج	Diastolic pressure in both ventricles are equal.
د	MAP in small circulation is about ۱۴ mm Hg.
منبع:	

۵۵	۴۵. About the distribution of blood in the circulation choose the <u>incorrect</u> sentence.
الف	۴% of blood is in the heart
ب	۱۵% of blood is in the veins
ج	Capillaries have ۵% of blood
د	۱۹% of blood is in respiratory system
منبع:	

۵۶	۴۶. In the blood pressure issue, which one is correct?
الف	Static pressure increases with velocity increase
ب	Dynamic pressure is the same as filling pressure
ج	Total pressure is the sum of static and dynamic pressures
د	Dynamic pressure and static pressure are equal
منبع:	

۴۷. In the capillary issue select the right sentence.	۵۷
Capillaries are ۲ th division of aorta	الف
Intra-cellular oncotic pressure is ۲۸ mm Hg	ب
Lymphatic system is responsible for intracellular oncotic pressure	ج
In each minute there is ۲ ml extra fluid from capillary filtration	د
منبع:	

۴۸. The amount of pressure and cardiac output in left ventricle is which proportion of right ventricle respectively.	۵۸
۲۵% and ۵۰%	الف
۵۰% and ۵۰%	ب
۲۰% and ۱۰۰%	ج
۱۰۰% and ۵۰%	د
منبع:	

۴۹. A blood vessel resistance increases with....	۵۹
sympathetic stimulation	الف
decrease in its length	ب
decrease in viscosity	ج
increase in its radius	د
منبع:	

۵۰. In the heart issue which sentence is <u>not</u> correct?	۶۰
Ventricular highest blood volume is in contraction with constant volume time	الف
Ventricular lowest blood volume is in dilatation with constant volume time	ب
All ۴ valves are closed in both constant volumes.	ج
Aortic valves opens in ۱۲۰ mm/hg and closes in ۸۰ mm Hg	د
منبع:	

۵۱. Mean arterial flow in a person is ۷۰ mm/ Hg how are the baroreceptor and chemoreceptor activity?	۶۱
They don't have any activity	الف
Baroreceptors are active and chemoreceptor inactive	ب
Only chemoreceptors are active and baroreceptor are inactive	ج
Both receptors are active with low frequency	د
منبع:	

۵۲. Which of the followings is not the homeostatic action of blood circulation?		۶۲
Temperature regulation	الف	
Blood pressure	ب	
Cells immunity	ج	
Feeding the cells	د	
منبع:		

۵۳. In blood flow issue which sentence is not correct?		۶۳
Reduction in RBC No. reduces critical closure point unit.	الف	
Decrease in filling pressure will happen with vasodilatation.	ب	
With increase in the artery diameter the pressure to the wall would increase.	ج	
An increase in blood flow, the viscosity effect will decrease	د	
منبع:		

۵۴. In which of the following vessels the pulse pressure is near zero?		۶۴
Femoral artery	الف	
Radial artery	ب	
Arterioles	ج	
Capillaries	د	
منبع:		

۵۵. In Venus return which one is correct?		۶۵
In the laying down position vein valves have main role	الف	
In the laying down position hydrostatic pressure is near to ۹۰ mm/hg	ب	
Misfunction of vein valve is main reason for varicose vein	ج	
In the upright position blood pressure is the most important factor for venus return	د	
منبع:		

۵۶. Which of the following organs have highest blood flow per minute and per ۱۰۰ gram of tissue respectively?		۶۶
Kidney- GI	الف	
Liver- kidney	ب	
Brain- bones	ج	
GI- liver	د	
منبع:		

		۶۷
۵۷. Which of the following is not correct?		
Increase in Ca level cause vasoconstriction	الف	
Increase in K ion in normal range cause vasodilation	ب	
Increase in Mg level cause vasoconstriction	ج	
Increase in H level cause vasodilation	د	
		منبع:

		۶۸
۵۸. Which of the following is not correct?		
Baro receptor activates from pressure ۶۰	الف	
Chemo receptor activates when pressure comes below ۸۰	ب	
Baro receptors are responsible for fine tuning of blood pressure	ج	
Chemo receptors are responsible for Bainbridge reflex	د	
		منبع:

		anatomy	Question time (seconds): →
۶۹	At the opening of which of the following vessels in heart valve is seen?		
	۱	left pulmonary veins	
	۲	Vena cava Superior	
	۳	Right pulmonary veins	
	۴	Coronary sinus	
Source:			
Description:			

		Question time (seconds): →	
۷۰	The phrenic nerve innervates all of the following areas except...?		
	۱	Diaphragm muscle	
	۲	Visceral serous pericardium	
	۳	Parietal serous pericardium	
	۴	Fibrous pericardium	
Source:			
Description:			

			Question time (seconds): →
۷۱	Which rib is the most curved rib?		
	۱	First	
	۲	Second	
	۳	eleventh	
	۴	۱۲th	
Source:			
Description:			

			Question time (seconds): →
۷۲	Which of the following joints is cartilaginous?		
	۱	First Costo transverse	
	۲	First costovertebral	
	۳	Second costo-sternal	
	۴	Intervertebral disc	
Source:			
Description:			

			Question time (seconds): →
۷۳	Which of the following muscles is connected to the ribs?		
	۱	Trapezius	
	۲	Serratus anterior	
	۳	Rhomboid minor	
	۴	Rhomboid major	
Source:			
Description:			

		Question time (seconds): →	
۷۴	All the following veins drain into the coronary sinus except?		
	۱	Middle cardiac vein	
	۲	Great cardiac vein	
	۳	Oblique cardiac vein	
	۴	Vena cava inferior	
Source:			
Description:			

		Question time (seconds): →	
۷۵	Which of the following muscles raise the ribs?		
	۱	Inferior posterior serratus	
	۲	Intima	
	۳	External intercostal	
	۴	Transversal thoracic	
Source:			
Description:			

		Question time (seconds): →	
۷۶	All the following structures are seen in the left ventricle except...		
	۱	Infundibulum	
	۲	apex of the heart	
	۳	papillary muscles	
	۴	Chorda tendio	
Source:			
Description:			

		Question time (seconds): ➔
۷۷	The fibrous skeleton of the heart can be seen in all the following parts of the heart except...	
	۱	At the beginning of the aorta
	۲	At the beginning of the pulmonary trunk
	۳	Right atrioventricular orifice
	۴	Coronay sinus
Source:		
Description:		

		Question time (seconds): ➔
۷۸	All of the following elements pass through the back of the esophagus except... A) b) c) Subsidiary hemizygous vein d) channel	
	۱	Hemizygous vein
	۲	Thoracic duct
	۳	left vagus nerve
	۴	Accessory Hemizygous vein
Source:		
Description:		

		Question time (seconds): ➔
۷۹	Which of the following elements passes through the larg foramen of diaphragm ?	
	۱	Left phrenic nerve
	۲	Greater splancnic nerve
	۳	Azygos vein
	۴	Right phrenic nerve
Source:		
Description:		

			Question time (seconds): →
۸۰	All the following vessels are seen in the coronary sulcus except		
	۱	Left coronary artery	
	۲	Right coronary artery	
	۳	Coronary sinus	
	۴	Diagonal artery	
Source:			
Description:			

			Question time (seconds): →
۸۱	In the chest X-ray image, the shadow on the right side of the heart belongs to which heart chamber?		
	۱	Left ventricle	
	۲	Left atrium	
	۳	Right atrium	
	۴	Right ventricle	
Source:			
Description:			

			Question time (seconds): →
۸۲	In surface anatomy, the image of the aortic valve is in front of which of the following areas?		
	۱	The third left intercostal space	
	۲	The second right intercostal space	
	۳	Second right costal cartilage	
	۴	Fourth left costal cartilage	
Source:			
Description:			

		Question time (seconds): ➔
۸۳	The sixth left posterior intercostal vein drains into which of the following veins?	
	۱	Left internal thoracic
	۲	Hemiazygous
	۳	Accessory hemiazygous
	۴	Left brachiocephalic
Source:		
Description:		

		Question time (seconds): ➔
۸۴	Which of the following is true about the sympathetic chain?	
	۱	There is only gray ramus between segments T ^۱ to L ^۲
	۲	The ۵ th to ۹ th thoracic sympathetic ganglia are involved in the formation of the small splanchnic nerve
	۳	The white ramus is observed along the entire length of the sympathetic chain
	۴	The impar ganglion is seen in front of the coccyx
Source:		
Description:		

		Question time (seconds): ➔
۸۵	Which option is correct about the intercostal arteries?	
	۱	The right posterior intercostal artery is shorter than the left
	۲	The first left posterior intercostal artery branches from the thoracic aorta
	۳	The first anterior intercostal artery branches from the phrenic muscle
	۴	The anterior intercostal arteries are shorter than the posterior ones.
Source:		
Description:		

م			
		Question time (seconds): ➔	
۸۶	Which of the following vessels is behind the transverse pericardial sinus?		
	۱	Aortic artery	
	۲	Pulmonary trunk	
	۳	Vena cava Inferior	
	۴	Superior vena cava	
Source:			
Description:			

		Question time (seconds): ➔	
۸۷	All of the following are observed in the superior mediastinum except...		
	۱	Right vagus nerve	
	۲	Thoracic duct	
	۳	Left recurrent laryngeal nerve	
	۴	Hemizygous vein	
Source:			
Description:			

		Question time (seconds): ➔	
۸۸	Where is the superficial cardiac plexus?		
	۱	At the posterior surface of the aortic arch	
	۲	At the lower side of the aortic arch	
	۳	In posterior mediastinum	
	۴	In right atrium	
Source:			
Description:			

		Question time (seconds): →
۸۹	What vein passes through the posterior interventricular sulcus?	
۱	Coronary sinus	
۲	Lesser cardiac vein	
۳	Greater cardiac vein	
۴	middle cardiac vein	
Source:		
Description:		

		Question time (seconds): →
۹۰	Where is the apex of the heart located in surface anatomy?	
۱	In the ۴ th intercostal space on the left, ۸ to ۹ cm from the sternum	
۲	In the ۴ th intercostal space on the right, ۸ to ۹ cm from the sternum	
۳	In the ۳ th intercostal space on the left, ۸ to ۹ cm from the sternum	
۴	In the ۳ th intercostal space on the right, ۸ to ۹ cm from the sternum	
Source:		
Description:		

		Question time (seconds): →
۹۱	Which of the following option is correct?	
۱	SA node is seen in left atrium	
۲	AV node is find in right atrium	
۳	Septomarginal is seen in right atrium	
۴	Fossa oval is seen in interventricular septum	
Source:		
Description:		

		Question time (seconds): →	
۹۲	All of the following sentences are correct except...?		
	۱	Manubrium make joint with ۳th rib	
	۲	Manubrium make joint with body of sternum	
	۳	Body of sternum is shorter than manubrium	
	۴	Manubrium is find anterior to inferior mediastinum	
Source:			
Description:			

		Question time (seconds): →	
۹۳	Which of the following sentences is correct?		
	۱	Lateral pectoral nerve innervate pectoralis minor	
	۲	Pectoralis minor attached to sternum	
	۳	Left upper limb lymph drain to thoracic duct.	
	۴	Thoracic duct drain to right subclavian vein	
Source:			
Description:			

		Question time (seconds): →	
۹۴	The heart is derived from:		
	۱	splanchnic mesoderm	
	۲	somatic mesoderm	
	۳	septum transversum	
	۴	intermediate mesoderm	
Source:			
Description:			

			Question time (seconds): →
۹۵	The most superior part of the inferior vena cava is derived from:		
	۱	left vitelline vein	
	۲	right vitelline vein	
	۳	right umbilical vein	
	۴	left umbilical vein	
Source:			
Description:			

			Question time (seconds): →
۹۶	Transposition of the great vessels is most directly related to abnormal development of which of the following		
	۱	Truncus arteriosus	
	۲	ductus venosus	
	۳	endocardial cushion	
	۴	septum primum	
Source:			
Description:			

ن			
			Question time (seconds): →
۹۷	The embryonic branchial arch arteries give origin to all of the following postnatal structures EXCEPT:		
	۱	arch of aorta	
	۲	ligamentum arteriosum	
	۳	pulmonary arteries	
	۴	ascending aorta	
Source:			
Description:			

		Question time (seconds): ➔	
۹۸	Each of the following associations correct EXCEPT:		
	۱	right sinus horn - coronary sinus	
	۲	right vitelline vein - inferior vena cava	
	۳	right posterior cardinal vein - azygos vein	
	۴	dorsal mesocardium - transverse sinus of the pericardium	
Source:			
Description:			

		Question time (seconds): ➔	
۹۹	The Tetralogy of Fallot is characterized by the following features EXCEPT:		
	۱	stenosis of the pulmonary artery	
	۲	interventricular septal defect	
	۳	overriding aorta	
	۴	hypertrophy of the left ventricle	
Source:			
Description:			

		Question time (seconds): ➔	
۱۰۰	Closure of the foramen primum results from fusion of the:		
	۱	septum secundum and the fused endocardial cushions	
	۲	septum secundum and the septum primum	
	۳	septum primum and the fused endocardial cushions	
	۴	septum primum and the septum spurium	
Source:			
Description:			